STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-SPECIAL-GOOD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5631-5643 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 5645-5757, the complete author proof (111 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 0 of queue_sdga.json (pb=4279, stmt 5631-5643, proof 5645-5757).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FSG: sources/stacks-sdga/pinned-original-source.tex, source lines 3129–3137. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSC: sources/stacks-sdga/pinned-original-source.tex, source lines 2983–2991. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSD: sources/stacks-sdga/pinned-original-source.tex, source lines 3022–3029. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In the situation above the differential graded $\mathcal{O}$-algebra*

$$\mathcal{A} = \operatorname{colim} \mathcal{A}_i$$

has the following property: for any morphism $(f, f^\sharp) : (Sh(\mathcal{C}'), \mathcal{O}') \to (Sh(\mathcal{C}), \mathcal{O})$ of ringed topoi, the pullback $f^*\mathcal{A}$ is flat as a graded $\mathcal{O}'$-module and is K-flat as a complex of $\mathcal{O}'$-modules.

Proof. Observe that $f^*\mathcal{A} = \operatorname{colim} f^*\mathcal{A}_i$ and that

$$f^*\mathcal{A}_i = \mathcal{O}'\langle f^{-1}\mathcal{S}_0 \amalg \ldots \amalg f^{-1}\mathcal{S}_i\rangle$$

with differential given by the inductive procedure above using $f^{-1}\delta_{i+1}$. Thus it suffices to prove that $\mathcal{A}$ is flat as a graded $\mathcal{O}$-module and is K-flat as a complex of

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$\mathcal{O}$-modules. For this it suffices to prove that each $\mathcal{A}_i$ is flat as a graded $\mathcal{O}$-module and is K-flat as a complex of $\mathcal{O}$-modules, compare with Lemma 0FSD.

For $i \geq 1$ write $\mathcal{S} = \mathcal{S}_0 \amalg \ldots \amalg \mathcal{S}_i$ so that we have $\mathcal{A}_i = \mathcal{O}\langle\mathcal{S}\rangle$ as a graded $\mathcal{O}$-algebra. We are going to construct a filtration of this algebra by differential graded $\mathcal{O}$-submodules.

Set $W = \mathbf{Z}_{\geq 0}^{i+1}$ considered with lexicographical ordering. Namely, given $w = (w_0, \ldots w_i)$ and $w' = (w'_0, \ldots, w'_i)$ in W we say

$$w > w' \Leftrightarrow \exists j,\ 0 \leq j \leq i : w_i = w'_i,\ w_{i-1} = w'_{i-1},\ \ldots,\ w_{j+1} = w'_{j+1},\ w_j > w'_j$$

and so on. Suppose given a section $s = s_1 \cdot \ldots \cdot s_r$ of $\mathcal{S} \times \ldots \times \mathcal{S}$ over U. We say that the *weight of s is defined* if we have $s_a \in \mathcal{S}_{j_a}(U)$ for a unique $0 \leq j_a \leq i$. In this case we define the weight

$$w(s) = (w_0(s), \ldots, w_i(s)) \in W, \quad w_j(s) = |\{a \mid j_a = j\}|$$

The weight of any section of $\mathcal{S} \times \ldots \times \mathcal{S}$ is defined locally. The reader checks easily that we obtain a disjoint union decomposition

$$\mathcal{S} \times \ldots \times \mathcal{S} = \coprod\nolimits_{w \in W} (\mathcal{S} \times \ldots \times \mathcal{S})_w$$

into the subsheaves of sections of a given weight. Of course only $w \in W$ with $\sum_{0 \leq j \leq i} w_j = r$ show up for a given r. We correspondingly obtain a decomposition

$$\mathcal{A}_i = \mathcal{O} \oplus \bigoplus\nolimits_{r \geq 1} \bigoplus\nolimits_{w \in W} \mathcal{O}[(\mathcal{S} \times \ldots \times \mathcal{S})_w]$$

The rest of the proof relies on the following trivial observation: given r, w and local section $s = s_1 \cdot \ldots \cdot s_r$ of $(\mathcal{S} \times \ldots \times \mathcal{S})_w$ we have

$$\mathrm{d}(s) \text{ is a local section of } \mathcal{O} \oplus \bigoplus\nolimits_{r' \geq 1} \bigoplus\nolimits_{w' \in W,\ w' < w} \mathcal{O}[(\mathcal{S} \times \ldots \times \mathcal{S})_{w'}]$$

The reason is that in each of the expressions

$$(-1)^{\deg(s_1) + \ldots + \deg(s_{a-1})} s_1 \cdot \ldots s_{a-1} \cdot \delta(s_a) \cdot s_{a+1} \cdot \ldots \cdot s_r$$

whose sum give the element $\mathrm{d}(s)$ the element $\delta(s_a)$ is locally a $\mathcal{O}$-linear combination of elements $s'_1 \cdot \ldots \cdot s'_{r'}$ with $s'_{a'}$ in $\mathcal{S}_{j'_a}$ for some $0 \leq j'_{a'} < j_a$ where j_a is such that s_a is section of $\mathcal{S}_{j_a}$.

What this means is the following. Suppose for $w \in W$ we set

$$F_w\mathcal{A}_i = \mathcal{O} \oplus \bigoplus\nolimits_{r \geq 1} \bigoplus\nolimits_{w' \in W,\ w' \leq w} \mathcal{O}[(\mathcal{S} \times \ldots \times \mathcal{S})_{w'}]$$

By the observation above this is a differential graded $\mathcal{O}$-submodule. We get admissible short exact sequences

$$0 \to \mathrm{colim}_{w' < w} F_{w'}\mathcal{A}_i \to F_w\mathcal{A}_i \to \bigoplus\nolimits_{r \geq 1} \mathcal{O}[(\mathcal{S} \times \ldots \times \mathcal{S})_w] \to 0$$

of differential graded $\mathcal{A}$-modules where the differential on the right hand side is zero.

Now we finish the proof by transfinite induction over the ordered set W. The differential graded complex $F_0\mathcal{A}_0$ is the summand $\mathcal{O}$ and this is K-flat and graded flat. For $w \in W$ if the result is true for $F_{w'}\mathcal{A}_i$ for $w' < w$, then by Lemmas 0FSD, 0FSC, and 0FSG we obtain the result for w. Finally, we have $\mathcal{A}_i = \mathrm{colim}_{w \in W} F_w\mathcal{A}_i$ and we conclude. $\square$